FY-926

FIRST YEAR HIGHER SECONDARY
IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

BIOLOGY
(Botany & Zoology)

Maximum : 60 Scores

Time : 2 Hours
Cool-off time : 15 Minutes
Preparatory Time : 10 Minutes

General Instructions to Candidates :

- There is a 'Cool-off time' of 15 minutes in addition to the writing time. Further there is a '10 minutes' 'Preparatory Time' at the end of the Botany Examination and before the commencement of Zoology Examination.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

വിദ്യാർത്ഥികൾക്കുള്ള പൊതുനിർദ്ദേശങ്ങൾ :

- നിർദ്ദിഷ്ട സമയത്തിന് പുറമെ 15 മിനിറ്റ് 'കൂൾ ഓഫ് ടൈം' ഉണ്ടായിരിക്കും. കൂടാതെ ബോട്ടണി പരീക്ഷയ്ക്കുശേഷം സുവോളജി പരീക്ഷ തുടങ്ങുന്നതിനുമുമ്പ് '10 മിനിറ്റ്' തയ്യാറെടുപ്പുകൾ നടത്തുന്നതിനായി നല്കുന്നതാണ്.
- ചോദ്യങ്ങൾ പരിചയപ്പെടാനും ഉത്തരങ്ങൾ ആസൂത്രണം ചെയ്യാനും 'കൂൾ ഓഫ് ടൈം' ഉപയോഗിക്കുക.
- ഉത്തരങ്ങൾ എഴുതുന്നതിനു മുമ്പ് ചോദ്യങ്ങൾ ശ്രദ്ധാപൂർവം വായിക്കണം.
- നിർദ്ദേശങ്ങൾ മുഴുവനും ശ്രദ്ധാപൂർവം വായിക്കണം.
- കണക്ക് കൂട്ടലുകൾ, ചിത്രങ്ങൾ, ഗ്രാഫുകൾ, എന്നിവ ഉത്തരപേപ്പറിൽ തന്നെ ഉണ്ടായിരിക്കണം.
- ചോദ്യങ്ങൾ മലയാളത്തിലും നല്കിയിട്ടുണ്ട്.
- ആവശ്യമുള്ള സ്ഥലത്ത് സമവാക്യങ്ങൾ കൊടുക്കണം.
- പ്രോഗ്രാമുകൾ ചെയ്യാനാകാത്ത കാൽക്കുലേറ്ററുകൾ ഒഴികെയുള്ള ഒരു ഇലക്ട്രോണിക് ഉപകരണവും പരീക്ഷാഹാളിൽ ഉപയോഗിക്കുവാൻ പാടില്ല.

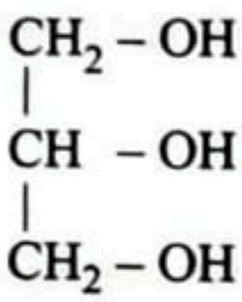
PART – B

ZOOLOGY

(Maximum : 30 Scores) **Time : 1 Hour**

I. Answer any 3 questions from 1 to 5. Each carries 1 score. **(3 × 1 = 3)**

1. Identify the biomolecule given below :



Answer : Glycerol

2. Expand ICZN. Answer : International Code of Zoological Nomenclature

3. Note the relationship between the first two words and find a suitable word for the fourth place.

RBC : Haemoglobin :: Muscles : Answer : Myoglobin

4. Name the centre present in the medulla region of brain that is primarily responsible for the regulation of respiration.

Answer : Respiratory Rhythm Centre

5. Name the hormone secreted by the heart wall which decreases blood pressure.

Answer : ANF/ Atrial Natri uretic factor

II. Answer any 9 questions from 6 to 16. Each carries 2 scores. **(9 × 2 = 18)**

6. Match the following :

A	B
Platyhelminthes	Pinnae
Birds	Radula
Mollusca	Flame cells
Mammals	Parapodia
	Pneumatic bones

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7. Categorise the following terms into two columns as primary and secondary metabolites :

Amino acids, Flavonoids, Rubber, Sugars

Primary Metabolite	Secondary Metabolite
Amino acid	Flavonoids
Sugars	Rubber

8. (a) Name the three main plasma proteins found in human blood.

(b) Plasma without clotting factor is called

(a) Fibrinogen, Globulins

(b) Serum

9. (a) Name the common chamber into which the digestive, reproductive, and excretory systems open in a frog.

(b) Write any one distinguishing feature between a male and a female frog.

(a) Cloaca

(b)

Male Frog	Female Frog
• Have vocal sacs • Have copulatory pad	• Lacks vocal sacs • Lacks copulatory pad

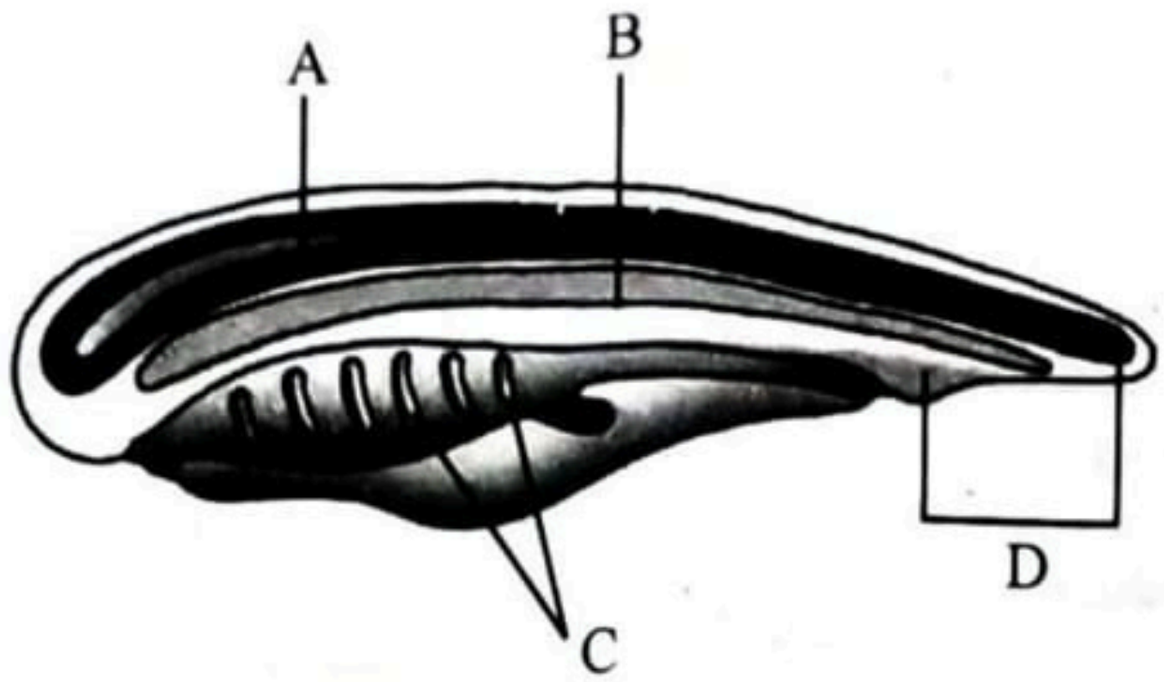
10. A specialised cardiac musculature called nodal tissue is distributed in the heart.

Prepare a flow chart showing the pathway of cardiac impulses through the following nodal tissues :

AVN, Purkinje fibres, AV bundle, SAN

SAN → AVN → AV Bundle → Purkinje fibres

11. Label the chordate characters A, B, C and D shown in the diagram :



A- Nerve cord
B- Noto chords
C- Gill slits
D- Post anal tail

12. The non-protein part of an enzyme is called a cofactor. There are three types of cofactors : prosthetic groups, coenzymes and metal ions.

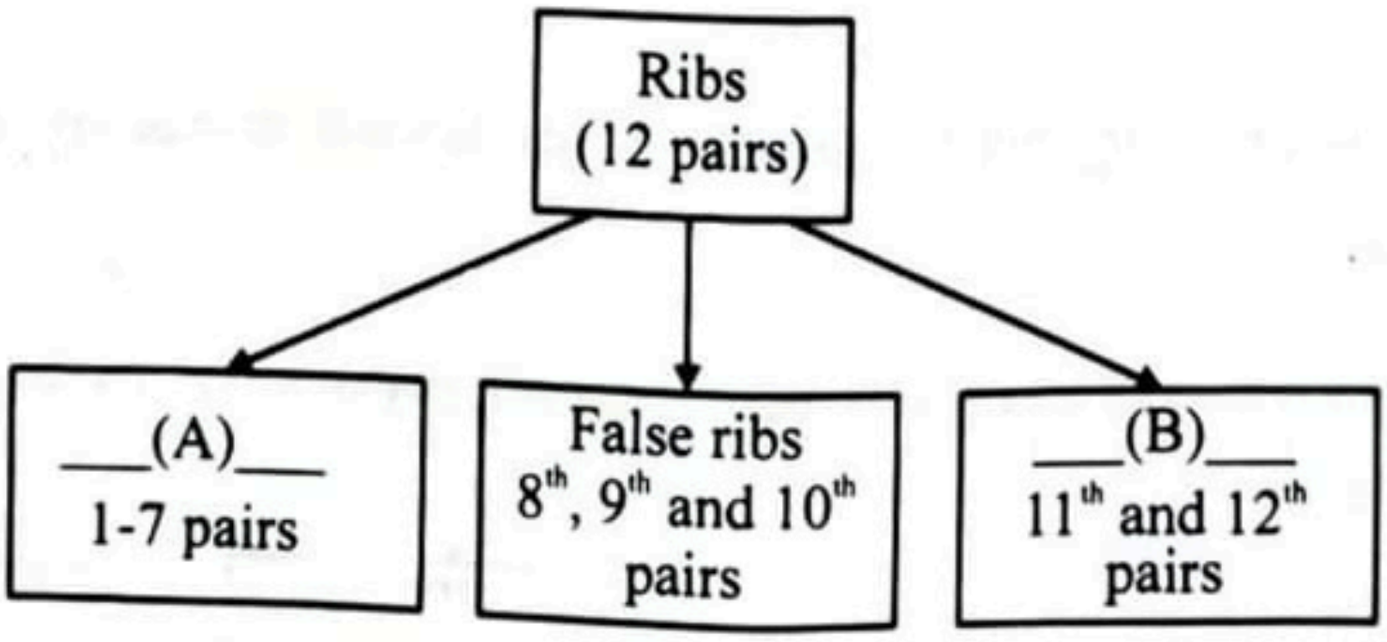
(a) Which type of cofactor is tightly bound to the enzyme ?

(b) The protein part of an enzyme is called apo.

(c) What happens when cofactor is removed from the enzyme ?

(a) Prosthetic groups
(b) The protein part of an enzyme is called an apoenzyme.
(c) When the cofactor is removed from an enzyme, its catalytic activity is lost, rendering the enzyme completely inactive.

13. (a) Complete the chart showing the different types of ribs :



(b) Inflammation of joints due to the accumulation of uric acid crystals is called _____.

(a) A- True ribs
B- Floating ribs
(b) Gout

14. In ___(A)___ type of synapse, the membranes of the presynaptic and postsynaptic neurons are in close proximity.

In ___(B)___ type of synapse, the membranes of the presynaptic and postsynaptic neurons are separated by a fluid-filled space.

(a) Identify the types of synapses (A) and (B).

(b) Name the fluid-filled space present in synapse (B).

(a) A- Electrical synapse
B- Chemical synapse
(b) Synaptic cleft

15.



(a) Identify the animal.

(b) Name the class to which it belongs.

(a) Sea horse
(b) Osteichthyes

16. **Which hormonal deficiency is responsible for the following diseases ?**

(a) **Diabetes mellitus**

(b) **Goitre**

(c) **Diabetes insipidus**

(d) **Dwarfism**

(a) Insulin
(b) Thyroxine
(c)) ADH / Vasopressin
(d) Growth hormone/ somatotropin

III. Answer any 3 questions from 17 to 20. Each carries 3 scores. **(3 × 3 = 9)**

17. Observe the picture and answer the following questions :



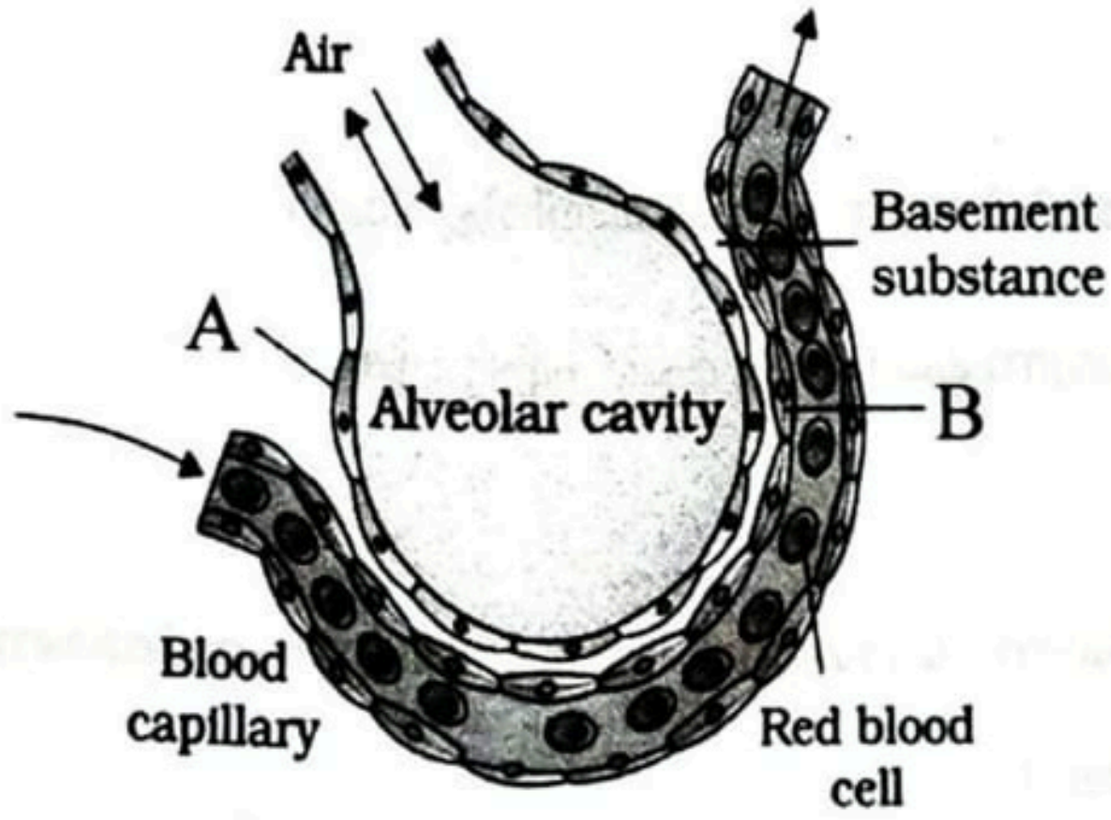
(a) **Identify the picture.**

(b) **How does this structure help the organism ?**

(c) **Name the phylum to which it belongs.**

(a) Cnidoblast
(b) - Defense against predators
- Capture of prey (offense)
- Anchorage to surfaces
(c) It belongs to the phylum Cnidaria (Coelenterata).

18. Observe the diagram showing a section of an alveolus with a pulmonary capillary and answer the following questions :



(a) Label the diffusion membranes A and B.

(b) Write any two factors that affect the rate of diffusion across the diffusion membranes.

(c) What is the maximum number of oxygen molecules that each haemoglobin molecule can carry ?

(a) A: Squamous epithelium of the alveolar wall
B: Endothelium of the blood capillary
(b) - Solubility of the gases.
- Thickness of the diffusion membrane.
(c) 4 molecules of oxygen

19. (a) Which are the three main processes involved in urine formation ?

(b) Differentiate between cortical nephrons and juxta medullary nephrons.

(c) Inflammation of glomeruli of the kidney is called_____.

(a)

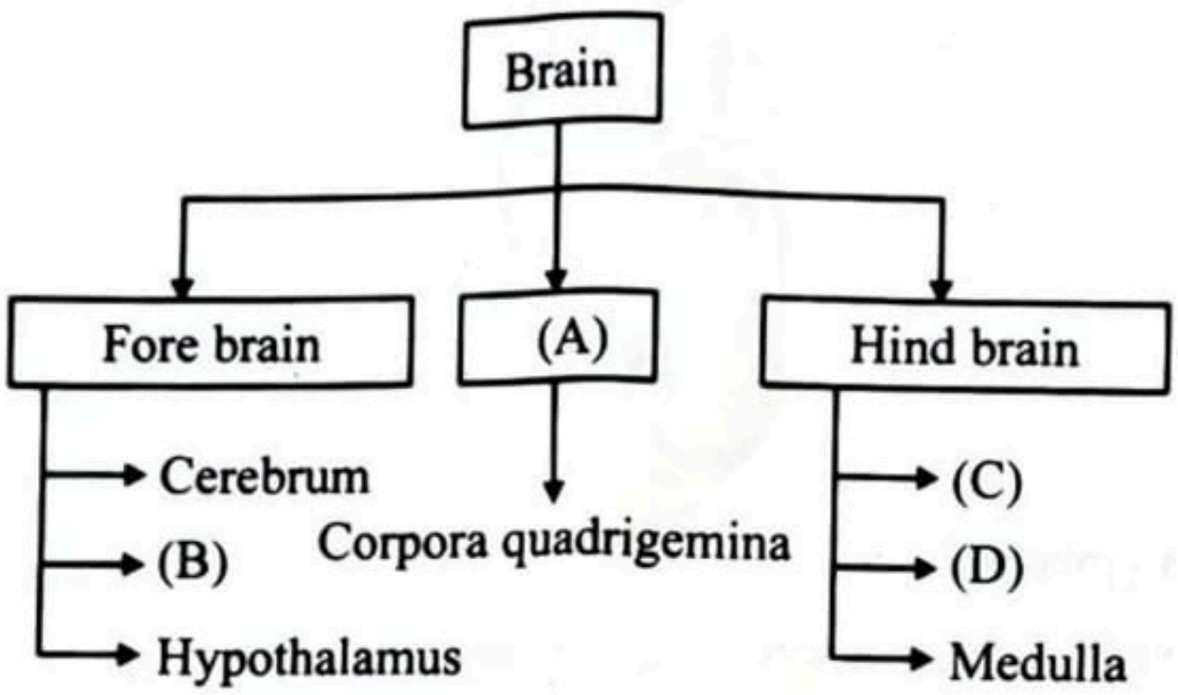
- **Glomerular filtration (Ultrafiltration)**
- **Selective reabsorption**
- **Tubular secretion**

(b)

Cortical Nephrons	Juxtamedullary Nephrons
Majority of our nephrones are cortical nephrones	Juxtamedullary Nephrons are few in number
Loop of Henle Very short and extends only a little into the medulla.	Very long and runs deep into the medulla.
Vasa Recta Absent or highly reduced.	Well-developed and prominent vasa recta.

(c) Glomerulonephritis.

20. (a) Complete the chart :



(b) The two cerebral hemispheres are connected by a tract of nerve fibres called _____.

(a) A- Mid Brain B- Thalamus
C- Cerebrum/pons D- Cerebellum

(b) Corpus callosum